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DFSMSHsm Best Practices

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Agenda

- Control Data Sets
- Recall
- Migration
- Audit
- Recycle
- Tape
- Throughput
- Miscellaneous

Record Level Sharing

- To improve overall DFSMSHsm performance, access the CDSs using **Record Level Sharing (RLS)**
- Some customers have reported significant performance improvements after switching to RLS
- Actual customer data, Bank 1, comparing nonRLS and RLS, with 1 yr elapsed:

Function	Increase in GBytes moved	Decrease in Window size
Auto Backup	33%	-25%
Migrate -> ML2	18%	-36%

- Actual customer, Bank 2, AUDIT before and after:
 - Before: Couldn't complete in 24 hrs
 - After: Complete within 4 hrs
- If you tried it and didn't see an improvement, it is most likely a configuration problem

GRS Star

- *Internal* performance testing has shown a significant improvement in CDS I/O intensive functions when using **GRS Star** as opposed to **GRS Ring**
 - **GRS Star** – A parallel sysplex implementation of Global Resource Serialization
 - Resource name list is placed in the coupling facility so that any request for a resource can be resolved with a single interaction
 - **GRS Ring** – A resource request must be passed to every participating member of the sysplex (ring)

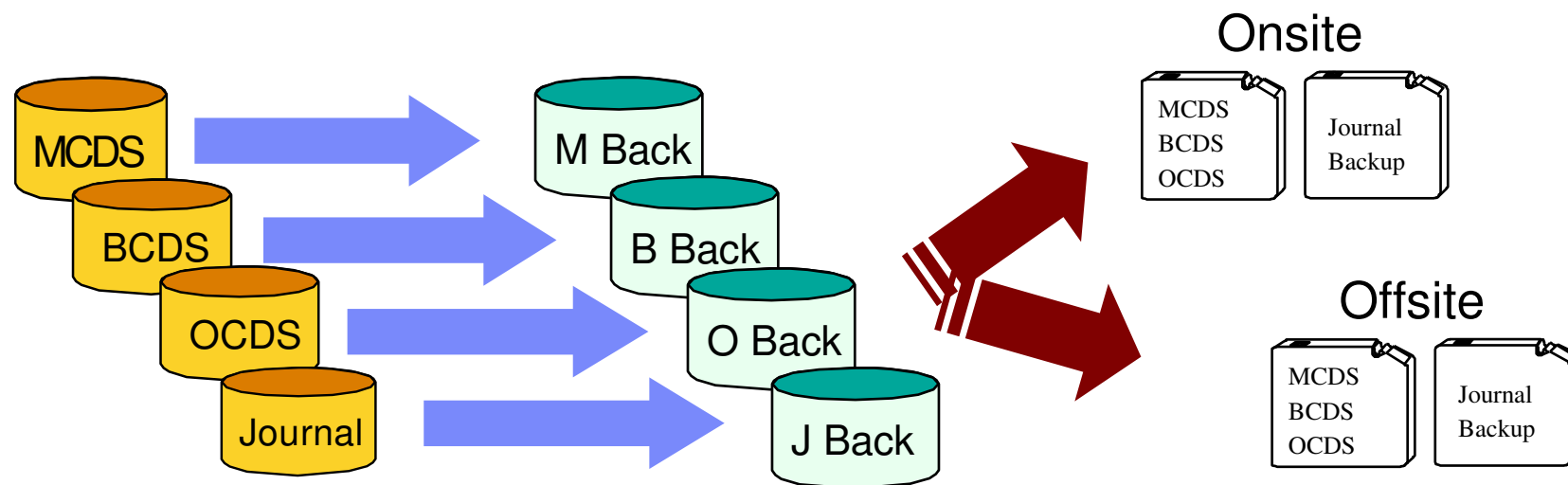
CDS Reorg

- Try to keep Reorganizing the CDSs to a minimum
- CDS Performance will be degraded for 2-3 weeks after a REORG
 - VSAM will perform a large number of CI / CA splits to create space for record insertions
 - Don't panic when HURBA / HARBA ratio increases during first few days
- Use FREESPACE(0 0) so that VSAM can create free space where it is needed
- ! Make sure all DFSMSHsm hosts in HSMplex are shutdown
 - This is one of the leading causes of breaking the CDSs
 - Use DISP=OLD in REORG job to prevent DFSMSHsm from starting

Duplex CDS Backups

- Create disk backup copies in parallel using PIT copy
- Use CB Exit to schedule a DFSMSDss dump job to create multiple copies of the disk backup copies

SETSYS EXITON(CB) CDSVERSIONBACKUP(DASD)



Health Checker / Backup

- Enable DFSMShsm Health Checker checks (OA19262)
 - **HSM_CDSB_BACKUP_COPIES**: Ensures that at least four CDS backup copies are being maintained
 - **HSM_CDSB_DASD_BACKUPS**: When backing up to disk, ensures that all CDS Backup copies exist
 - **HSM_CDSB_VALID_BACKUPS**: Determines if the number of *valid* backup copies has dropped below four

- Allocate the journal as a Large Format Sequential data set if you have to back up the CDSs more than once a day due to the journal filling up

CDS Recovery Planning

- Keep journal and disk backups separate from MCDS, BCDS and OCDS
- Minimize CDS Loss
 - Dual Copy / Remote Copy
 - Raid 5 or Raid 6
- Have documented and Tested CDS Recovery Plans
- Review “Data Recovery Scenarios” in DFSMSHsm Storage Administration Guide

Prioritize Recalls

- **RP Exit** can be used to prioritize data set Recall, Delete and Recover requests
- Priority range 0 – 100
 - Default priority is 50
- All Wait-type requests are prioritized higher than noWait-type requests
- Recall and Delete requests are on the same queue

DSN: CUSTMR.DS1
Priority: 100

DSN: PROD05.DS1
Priority: 80

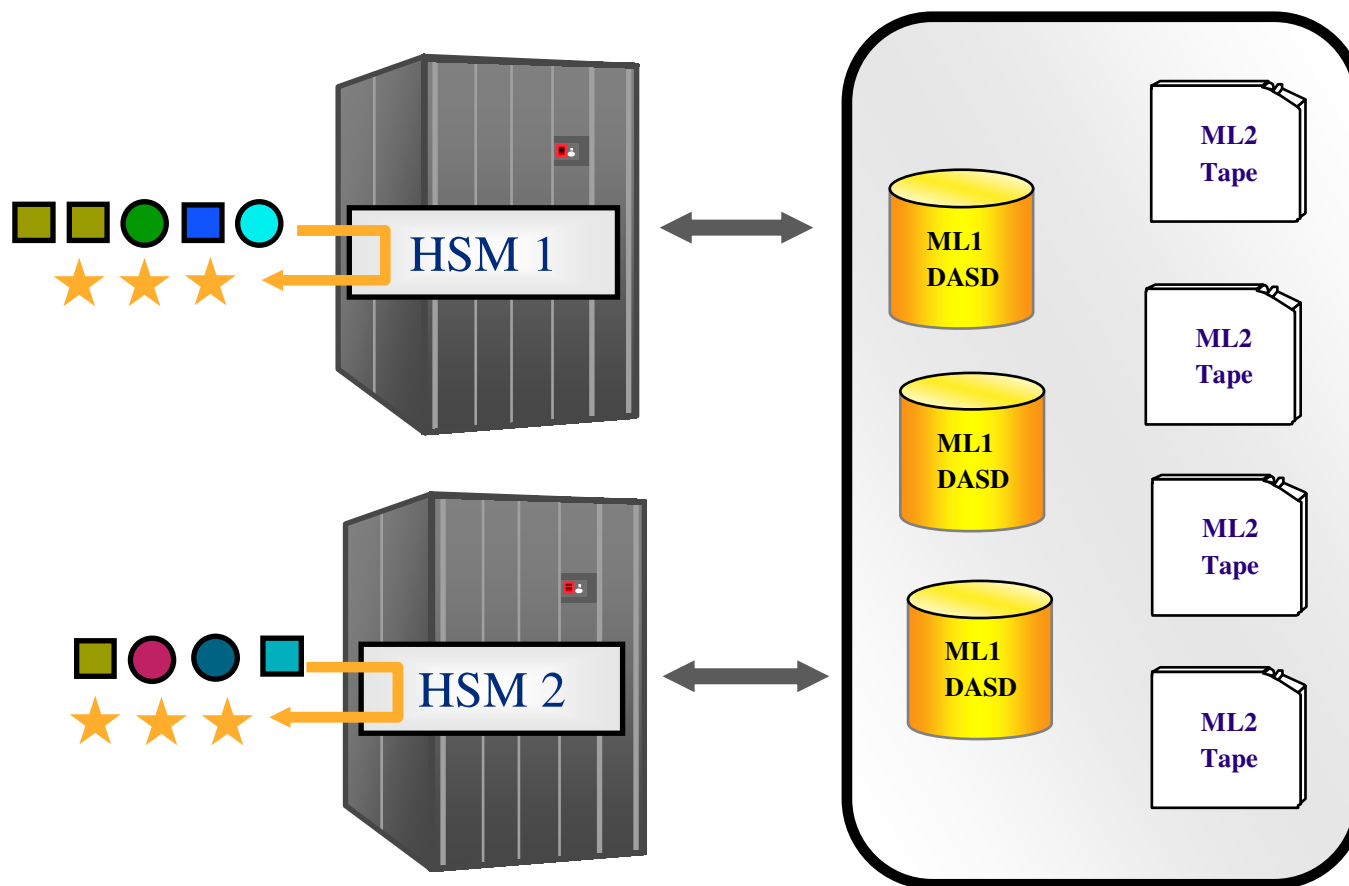
DSN: USERME.DS9
Priority: 50

DSN: UTIL01.DS2
Priority: 20

DSN: CLEANUP.DS7
Priority: 10

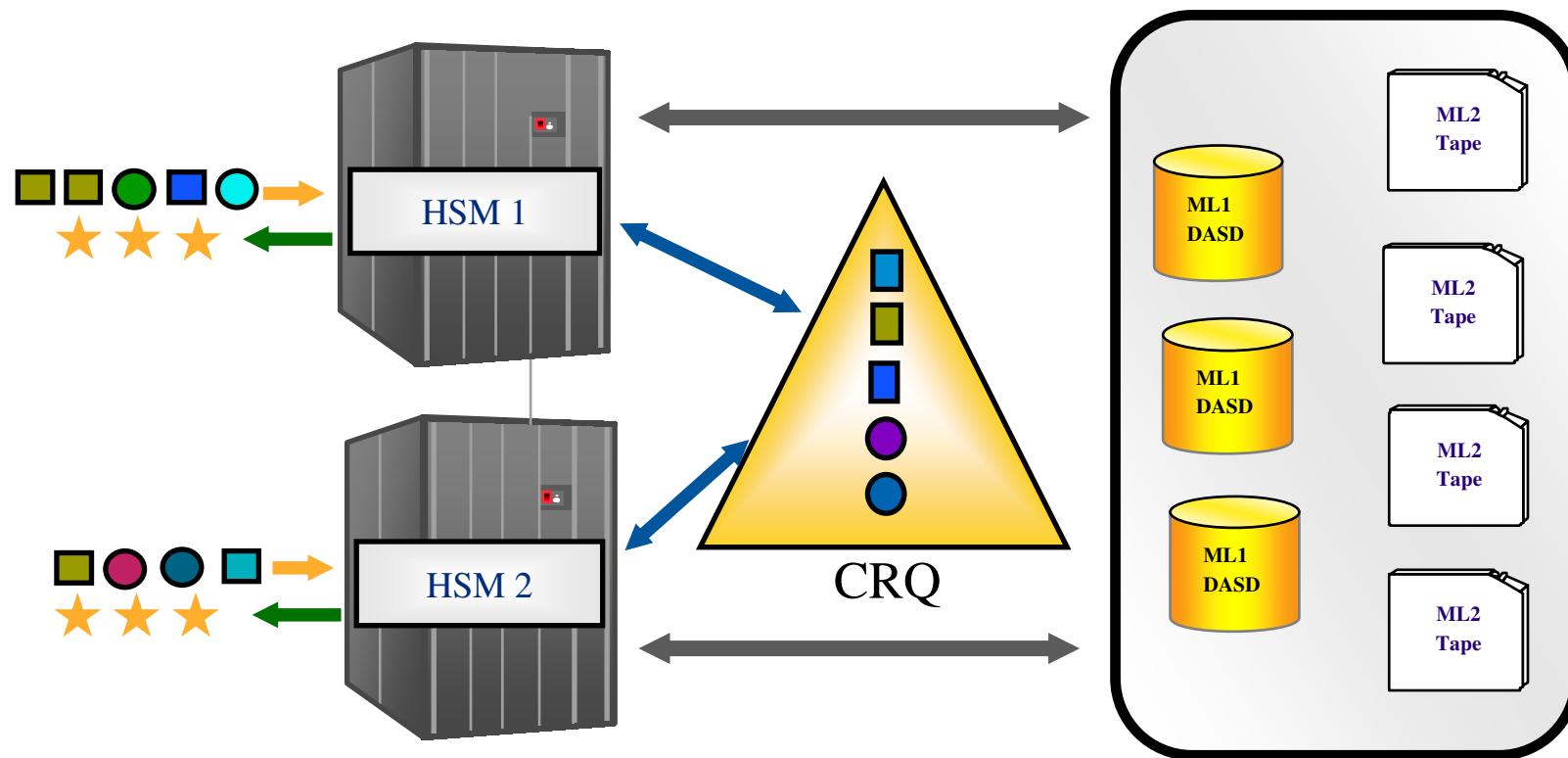
Common Recall Queue

- NonCRQ environment – Each host processes own requests



Common Recall Queue (cont)

- CRQ - All requests are placed onto a shared queue from which all hosts can select requests for processing
 - Implemented using a Coupling Facility List Structure



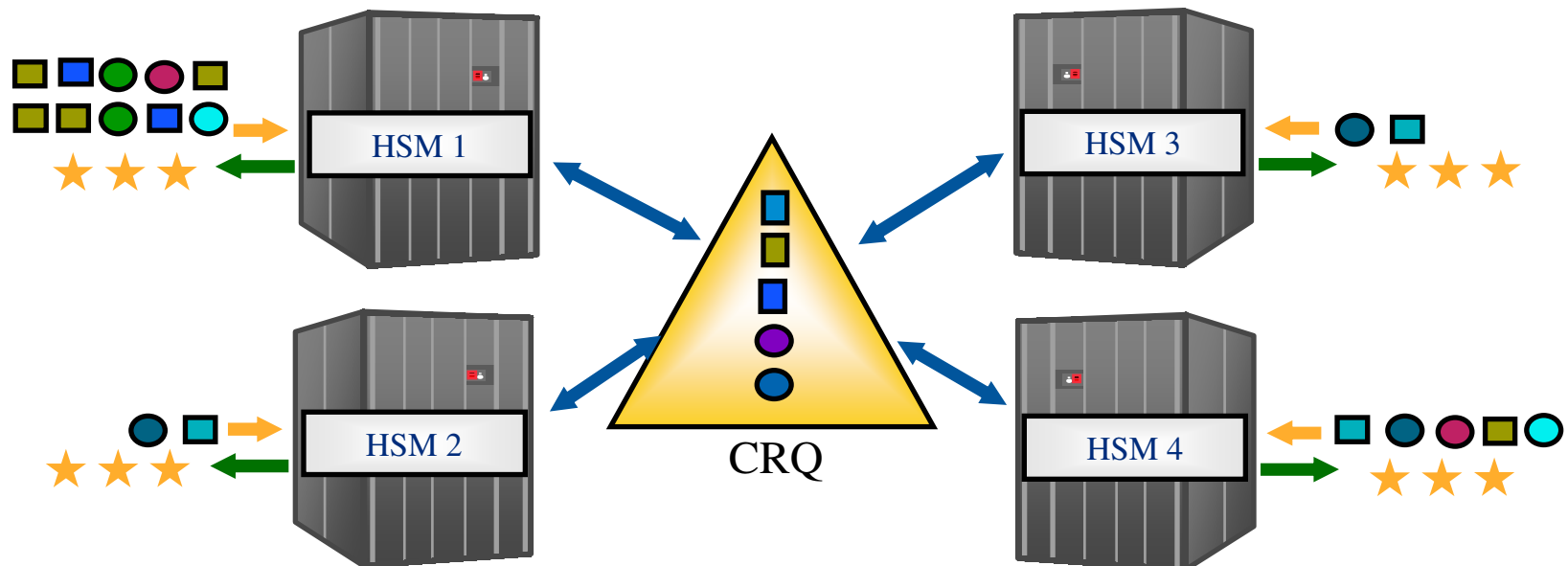
Common Recall Queue *(cont)*

- **Advantages of CRQ**
 - Workload balancing
 - Tape mount optimization
 - Quiesce Activity w/o impacting Recall
 - Priority optimization
 - Flexible configurations
 - Request persistence

Common Recall Queue (cont)

■ Workload balancing

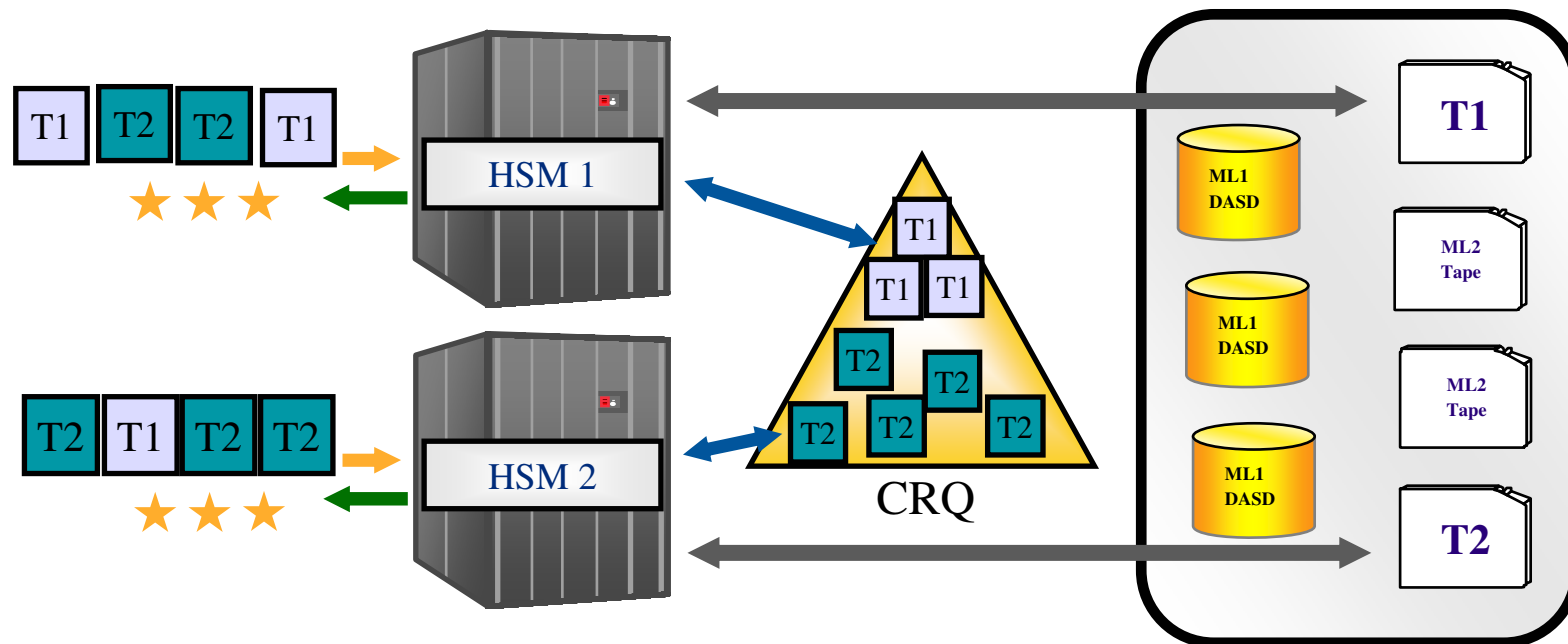
- Requests are evenly distributed among hosts until the maximum tasking level has been reached



Common Recall Queue (cont)

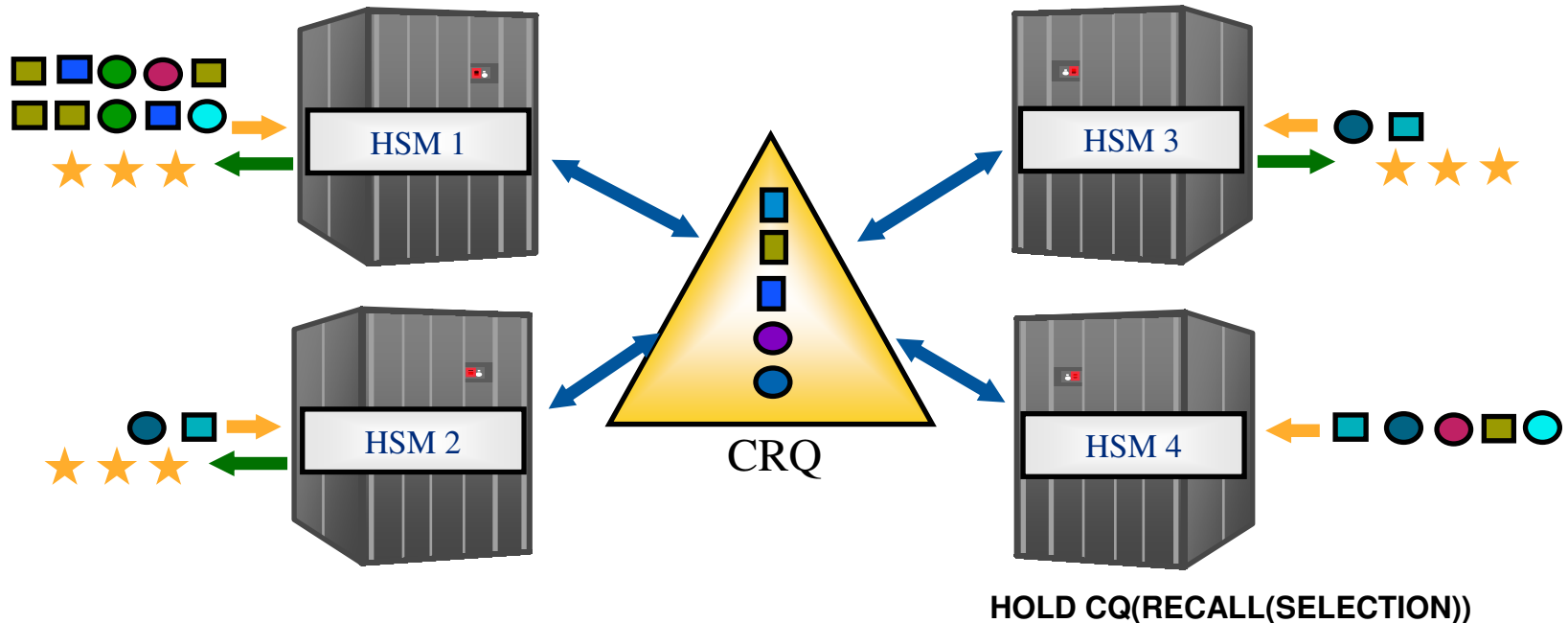
■ Tape Mount Optimization

- A recall task will process all requests in the CRQ that require the same tape
- ★ Only a single tape mount is required



Common Recall Queue (cont)

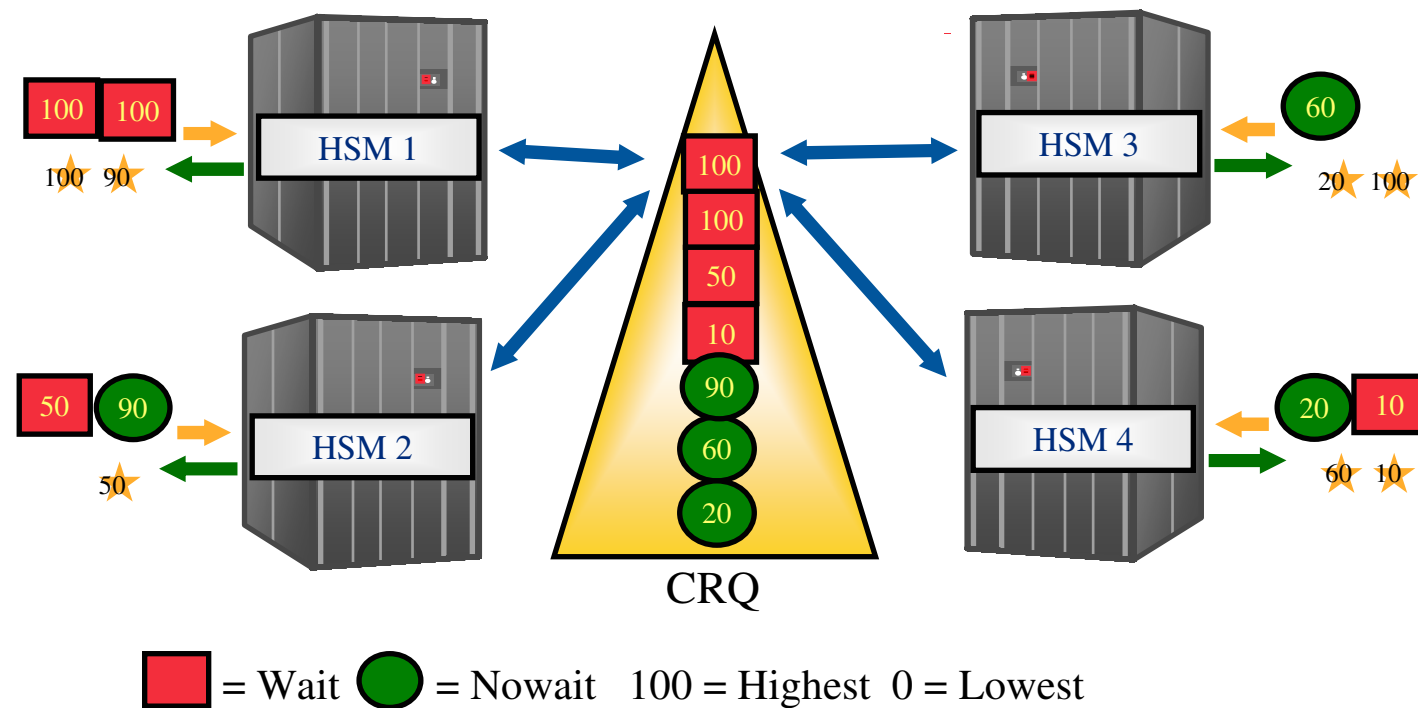
- Quiesce activity in preparation for a shutdown without holding Recall Activity
 - **HOLD CQ(RECALL(SELECTION))**
 - Places Recalls onto CRQ, but does not process any



Common Recall Queue (cont)

■ Priority Optimization

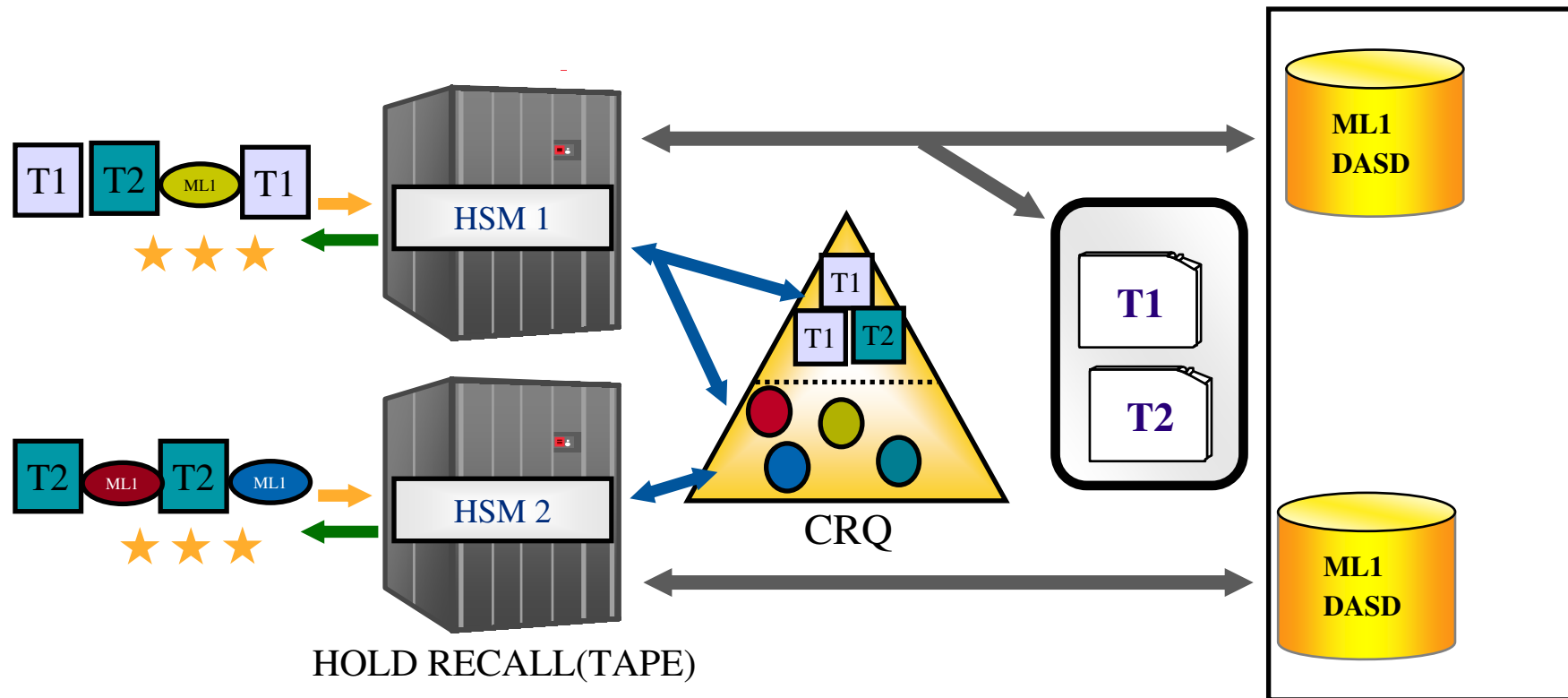
- Highest priority requests in the HSMplex are processed first



Common Recall Queue (cont)

Flexible Configurations

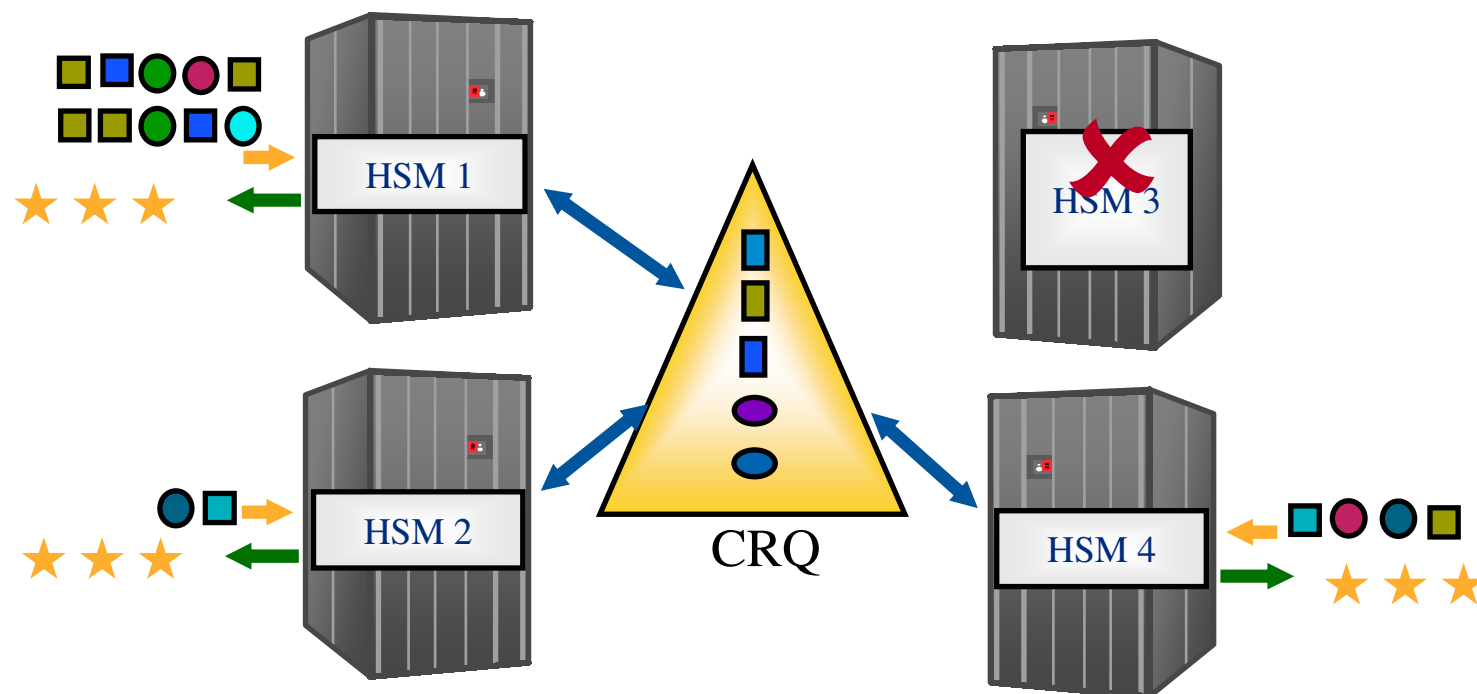
- Hosts not connected to tape drives can be configured to only select non-tape requests



Common Recall Queue (cont)

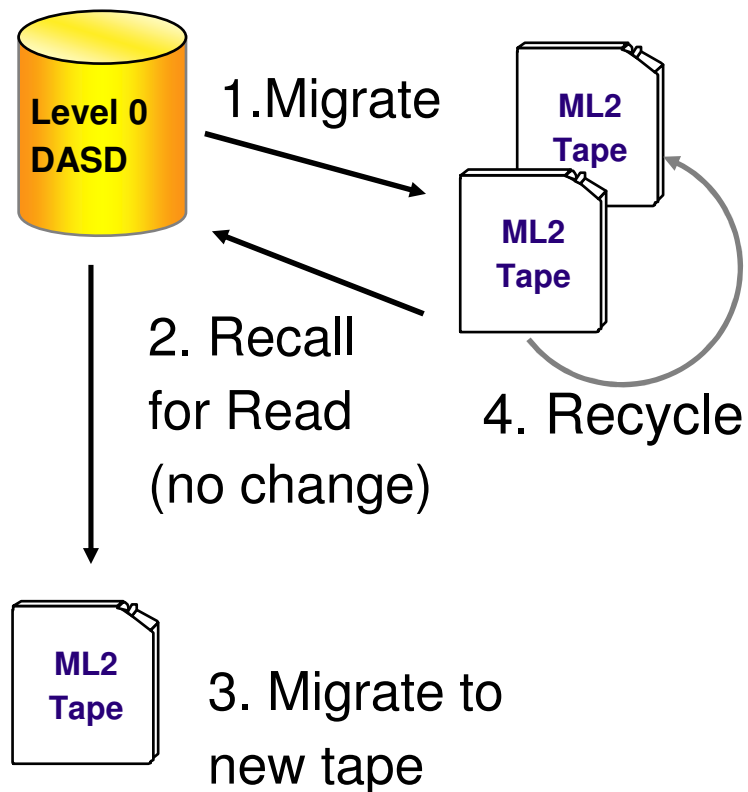
■ Request Persistence

- Outstanding Recall requests from unavailable hosts are processed by available hosts

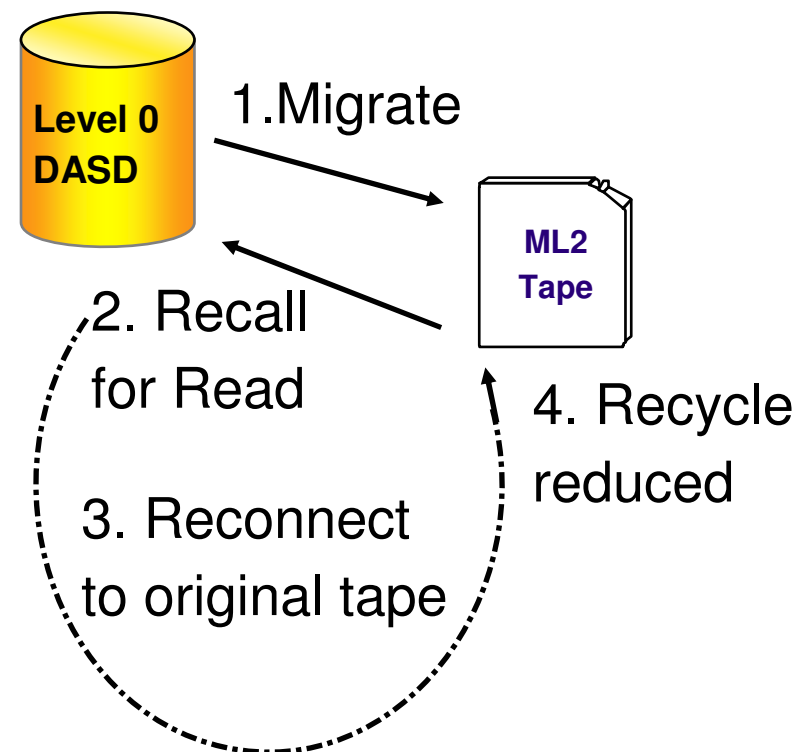


Fast Subsequent Migration

Without FSM



With FSM



Fast Subsequent Migration *(cont)*

★ Advantages

- No actual data movement for reconnection
 - Recycle work load reduced
 - SMS, nonSMS data sets supported
 - Reconnection done automatically and is transparent to user
-
- DFSMSHsm V1R7 updated this support to no longer rely on Data Set Change Indicator in VTOC to be OFF

SETSYS TAPEMIGRATION(RECONNECT(NONE | ALL | ML2DIRECT))

Duplex Tape Error Handling

SETSYS DUPLEX(MIGRATION(Y ERRORALTERNATE(CONT | MARKFULL)))

- For duplexing of migration tapes, both the original and the alternate will be marked full and two new tapes will be mounted
- ★ Guarantees that the original and alternate tapes are always identical
 - Greatly reduces the need for Tape Copies
 - No delay in creating the alternate copy
- Introduced in V1R8

Migration – Management Class Values

- In the Management class, pay attention to “Primary Days Non-usage” and “Level 1 Days Non-usage”
- “Level 1 Days Non-usage” includes time spent on Primary
- Example
 - Primary Days Non-usage = 4
 - Level 1 Days Non-usage = 4
- This results in data sets being migrated **directly to ML2** after 4 days
- In this case, “Level 1 Days Non-usage” should be ‘8’

Audit – Performance

- **OA13840** - Dramatically improves the performance of **AUDIT MEDIACONTROLS**
 - April 2006; All Releases
 - Requires IBM J70 tape control units with a microcode level of 1.19.3.5
 - Implements a *Hardware Assisted Search* to improve performance
 - Performance will vary depending on data set sizes on tape. Larger data set sizes get more benefit, smaller will get less.
 - CPU utilization is drastically reduced since fewer blocks are read from tape
 - If required tape CU and microcode not available, then legacy AUDIT code is used

Audit – Media Controls

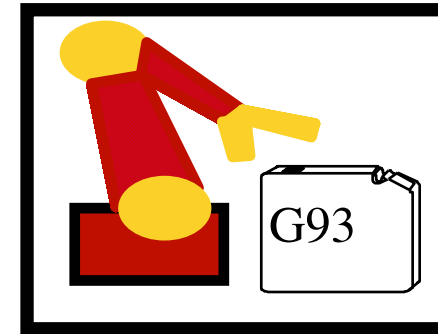
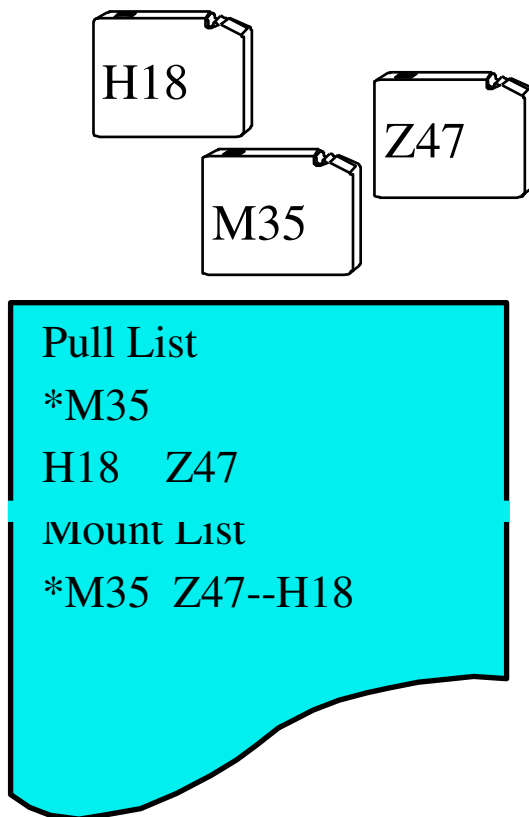
- **OA12976** - Enables Audit Mediacontrols to resume processing of migration or backup tape if:
 - AUDIT MEDCTL of a volume is held
 - DFSMSHsm is stopped
 - SETSYS EMERGENCY has been specified
- New **RESUME** parameter added to AUDIT MEDCTL VOLUMES(tape volser) FIX command
 - AUDIT cannot resume after ABENDS or I/O errors
- **RESUME** only valid when auditing a tape volume
- Valid only when **FIX** parameter is specified

AUDIT MEDCTL VOLUMES(A00342) RESUME FIX ODS('HSM.FIX')

Recycle – Limiting Workload

- Use the **LIMIT** parameter to match RECYCLE workload to your scratch tape needs:
 - LIMIT(50): Process enough input tapes to return a net gain of 50 scratch tapes
 - Example: Read 60 input, create 10 output
- Use **PERCENTVALID(0)** to reclaim empty tapes when no drives available

Recycle – Convert to New Tape Technologies



```
RECYCLE
SELECT(EXCLUDE(
  RANGE(G00:G99)))
TAPELIST EXECUTE
```

```
SETSYS RECYCLEOUTPUT(BACKUP |
  MIGRATION(unittype))
```

Recycle – Create Duplex Tapes

- Issue **SETSYS DUPLEX(MIGRATION(Y))** and/or **SETSYS DUPLEX(BACKUP(Y))** to begin duplexing migration and backup tapes
- Then Recycle all existing tapes
- You may want to specify **PERCENTVALID** to a high value on the generic recycle command if you want the duplexing to take effect more quickly!
- **TAPECOPY** is another method for creating duplex tapes but has the following limitations
 - Single threaded task structure
 - Is interruptible by recall, but if interrupted, must be restarted from beginning

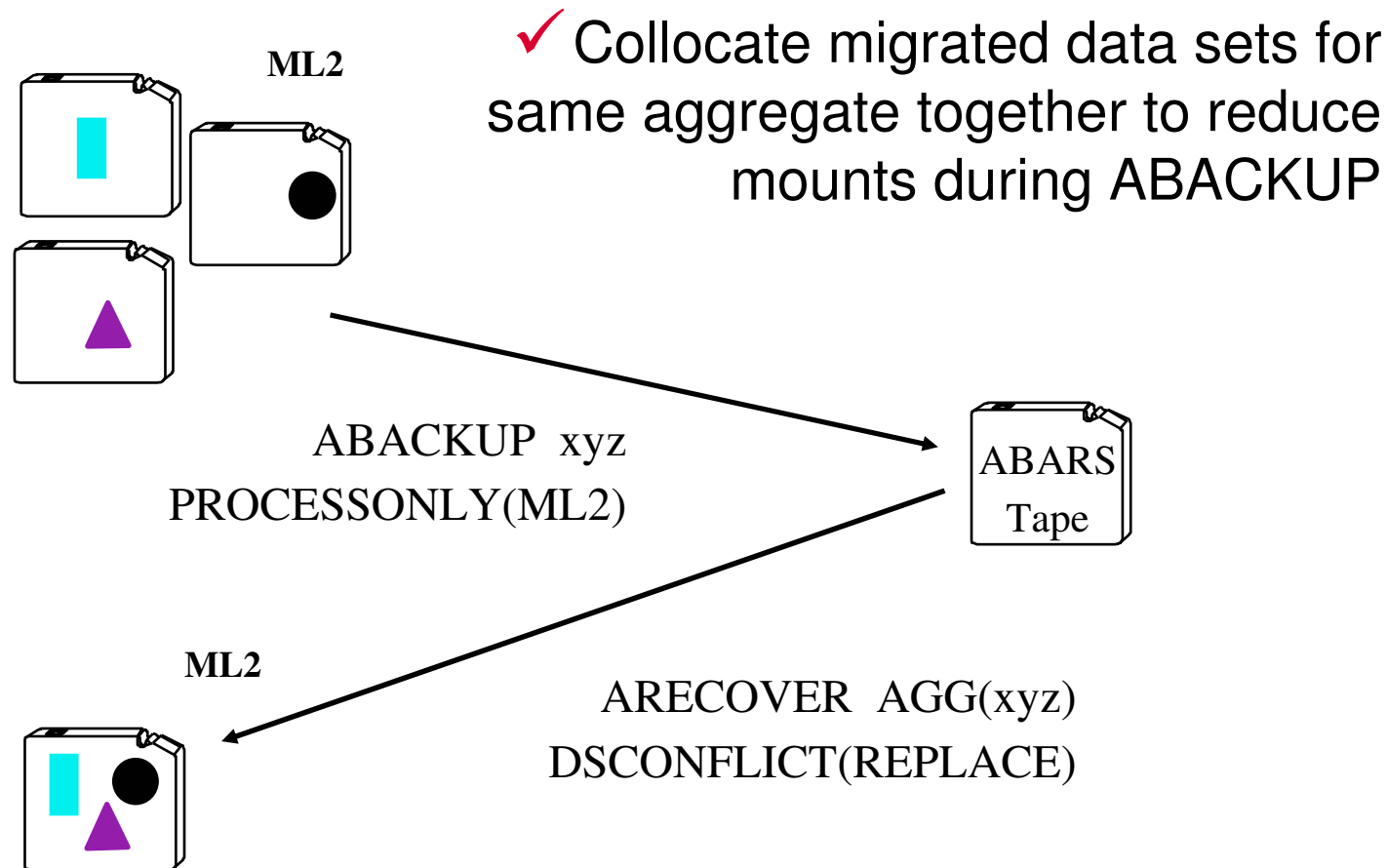
Tape – Connected Sets

- **Connected Set** - sequence of tape volumes connected by valid spanning data
 - Slows down recall and recycle activity
 - More difficult for tape library ejections
- 
- You can minimize the occurrence of connected sets with the judicious use of **SETSYS TAPEUTILIZATION(PERCENTFULL)** and **TAPESPANSIZE** parameters
 - **Never** use TAPEUTILIZATION(NOLIMIT)
 - For TAPESPANSIZE, a larger absolute value is needed to represent the same amount of unused capacity on a percentage basis when the tape has a larger total capacity. eg:
 - If you allow 2% of unused tape to avoid tape spanning for a 3590-Hxx device using enhanced media, you specify a TAPESPANSIZE of 1200 MB.
 - To allow 2% unused tape for a MEDIA5 tape on a 3592 Model J device (no performance scaling), you specify a TAPESPANSIZE of 6000 MB.
 - All size calculations for scaled tapes are based upon the scaled size and not the unscaled size.

Tape – Connected Sets *(cont)*

- You can determine if you have connected sets by issuing **LIST TTOC SELECT(CONNECTED)**
- Consider use of new **CHECKFIRST(N)** parameter on generic **RECYCLE** command if significant number of connected sets that meet PERCENTVALID criteria are not being recycled
- You can break a connected set by doing one of the following to the spanning data set
 - Recall a migrated data set
 - Deleting a data set backup using the BDELETE command

Tape – Collate ML2 for ABARS



DFSMSHsm and VTS

- In a VTS environment
 - SETSYS PARTIALTAPE should be MARKFULL, not REUSE
 - REUSE
 - Causes the complete virtual volume to be staged when DFSMSHsm reuses it
 - Perhaps worse yet, it causes a 'hole' in the physical tape from which the virtual tape came
 - MARKFULL
 - Only used portion of virtual volume de-staged to back-end tape
 - Does increase the number of virtual volumes required by DFSMSHsm

DFSMSHsm and VTS

- If using a 4GByte virtual volume in a VTS
 - You have to specify a multiplier value to inform DFSMSHsm that the capacity of the logical 3490 is greater than 400MB or 800MB.
 - **SETSYS TAPEUTILIZATION(LIBRARYBACKUP)** for backup
 - **SETSYS TAPEUTILIZATION(LIBRARYMIGRATION)** for migration
 - The multiplier will vary depending on whether the Data Class defines the DFSMSHsm backup tapes as CCST (400MB) or ECCST (800MB) capacity.
 - For example, a value of 997 is recommended for 400MB and 497 is recommended for 800MB
 - (1000 and 500 are not recommended because of tape spanning)

DFSMSHsm Fast Replication

- ★ Once a dump copy has started, DFSMSHsm prevents the target volumes from being overlaid with a new FlashCopy until all volumes have been dumped successfully
 - Overlaying the target volumes in the middle of the dump creates a data integrity exposure.
 - Once the target volumes have been overlaid, it is impossible to complete a dump copy!

DFSMSHsm prevents these!

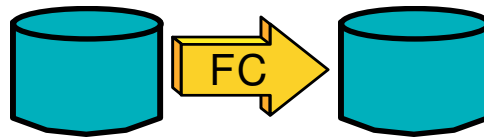
- Withdrawing a background copy while creating a dump copy creates a data integrity exposure

DFSMSHsm prevents this!

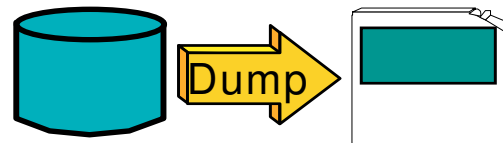
Fast Replication – Data Integrity

■ Scenario 1: Dump is in progress

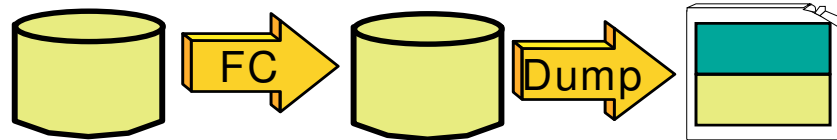
Time 1
Create Copy



Time 2
Start Dump



Time 3
Create new copy
before dump
completes



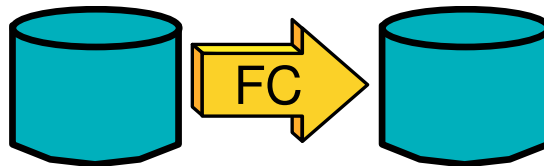
Tape is corrupt.
It represents a mix of
two points-in-time.

DFSMShsm prevents this!

Fast Replication – Data Integrity

■ Scenario 2: Dump fails

Time 1
Create Copy

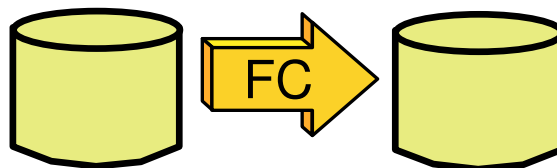


Time 2
Start Dump



Dump fails.

Time 3
Create new copy
before dump
retried.



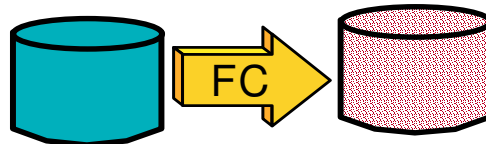
Chance to retry failed
dump is lost because
PIT copy was overlaid.

DFSMSHsm prevents this!

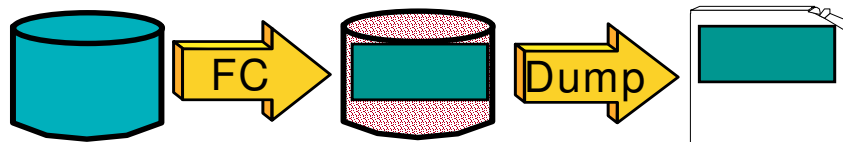
Fast Replication – Data Integrity

■ Scenario 3: Relationship is Withdrawn

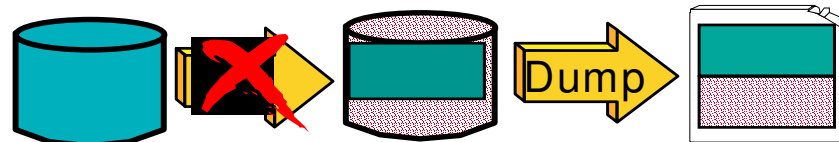
Time 1
Initiate FC



Time 2
Start Dump



Time 3
Withdraw
Relationship



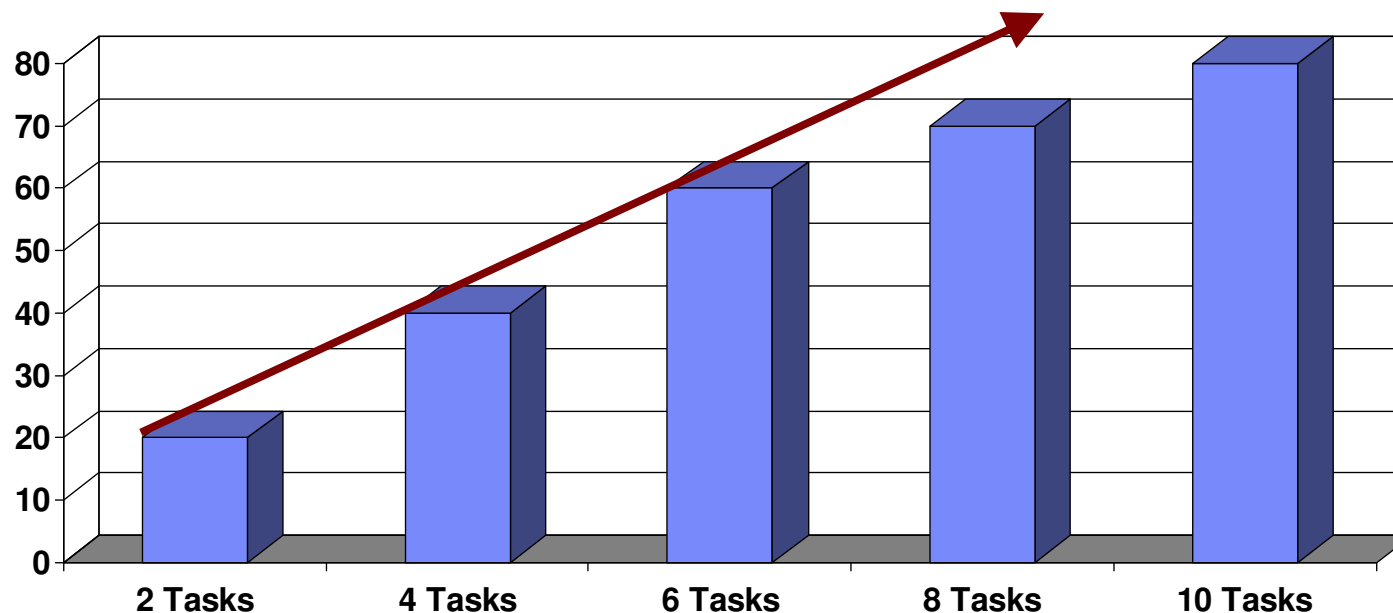
Tape is corrupt.
Data copied after the
withdraw is residual.

DFSMSHsm prevents this!

(When Withdraw done with DFSMSHsm)

Throughput - MASH

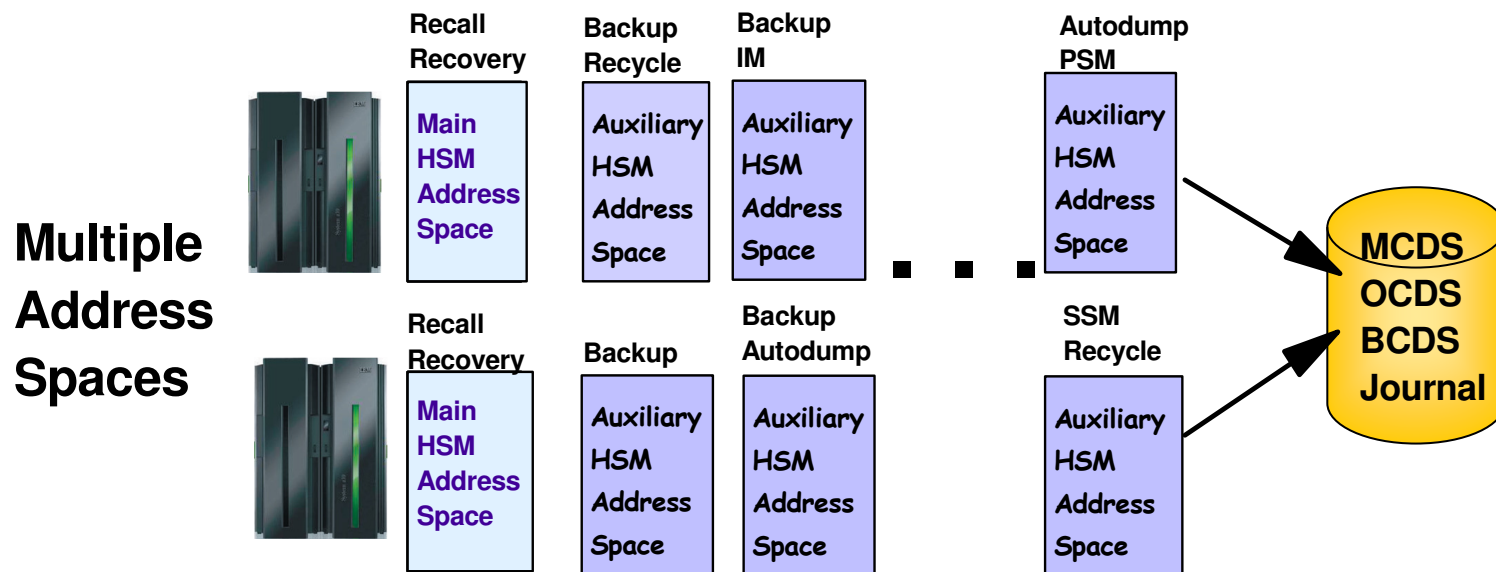
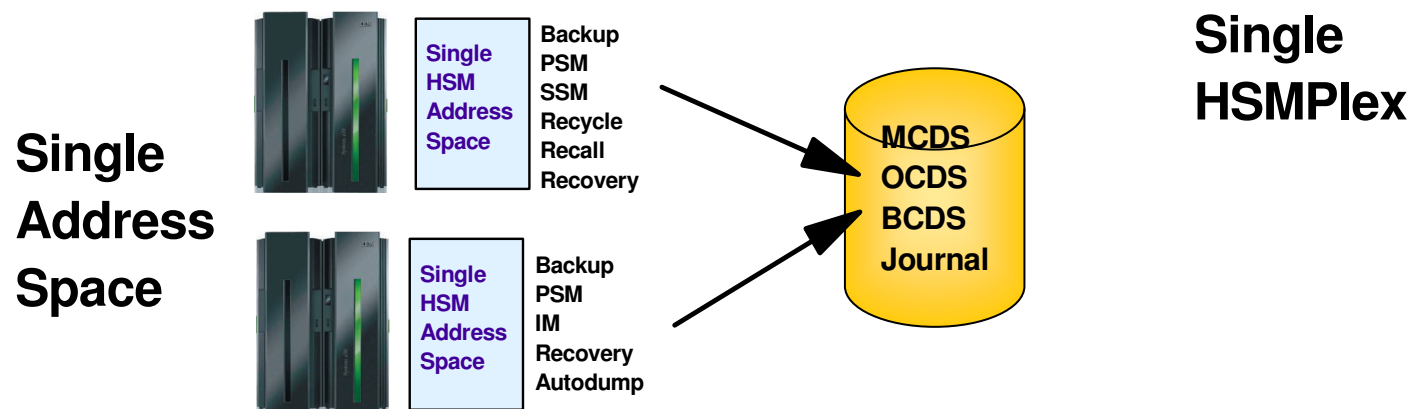
- In general, there is a performance 'knee' for DFSMSHsm functions
 - The knee for most functions is at 7-8 tasks
 - For Fast Replication the knee is at 24 of the possible 64 tasks
- Contention for the SYSZTIOT can be one cause



MASH *(cont)*

- Multiple Address Space HSM (MASH)
 - Each LPAR can have multiple active DFSMSHsm address spaces
 - Up to 39 active DFSMSHsms in an HSMplex
 - HSMplex – All DFSMSHsm's sharing the same control data sets
- Potential benefits of spreading out the DFSMSHsm workload to more hosts
 - Maintain tasks at optimal level
 - Increase overall tasking level
 - Hosts can be assigned different WLM Velocity Goals
 - Recall hosts via Common Recall Queue
 - Start hosts just to process Recall requests during high recall activity
 - Reduces SYSZTIOT contention for disk/tape allocations
 - Increased availability

MASH (cont)

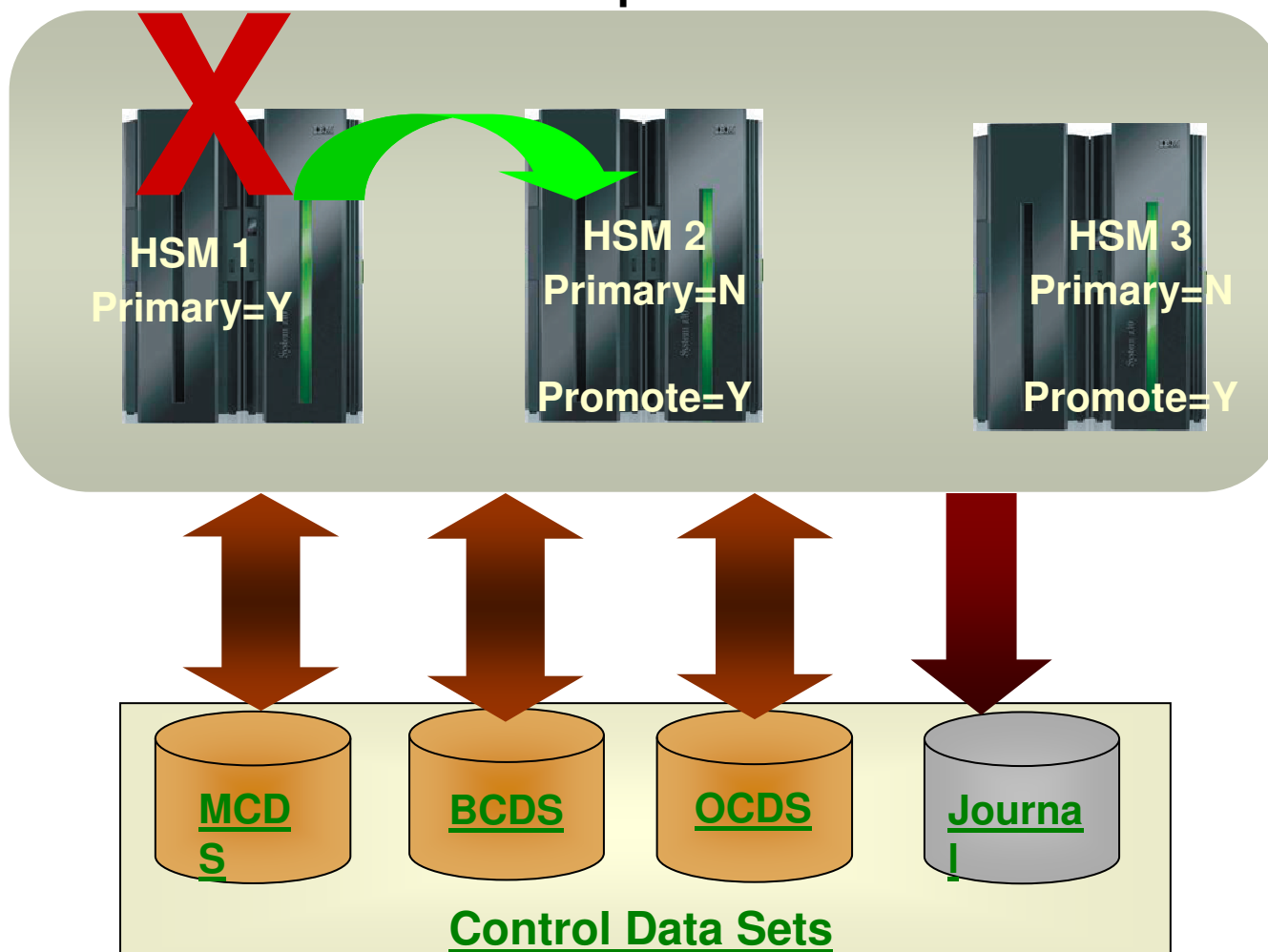


Secondary Host Promotion

- In an HSMplex, **Secondary Host Promotion** enables secondary DFSMSHsm hosts to take over the *unique* functions being performed by a disabled Primary and/or Secondary Space Management DFSMSHsm host
- A Primary DFSMSHsm host is the only host in an HSMplex that performs:
 - Hourly space checks
 - Automatic CDS backup
 - Automatic movement of backup versions from ML1 to tape
 - Automatic backup of migrated data sets
 - Expiration of dump copies
 - Deletion of excess dump VTOC copy data sets
- There is generally only a single DFSMSHsm host that performs SSM
- ! **Without SHP, when either the Primary or SSM host is disabled, the above functions are not performed**

Secondary Host Promotion

HSMplex



Secondary Host Promotion

DFSMSHsm host must be on a system within a HSMplex that has XCF active and configured in multisystem mode

- SETSYS PLEXNAME(HSMplex_name_suffix)
 - Default: ARCPLEX0
 - Must be specified if more than one HSMplex within a sysplex. Must be specified on all hosts in that HSMplex.
 - Must be specified in ARCCMDxx member
- SETSYS PROMOTE(PRIMARYHOST(Y|N) SSM(Y|N))
 - Default: No
 - PRIMARYHOST(Y) is ignored for Primary host
 - A SSM host cannot be promoted for another SSM host. ARC1521I issued if SSM(Y) specified on a SSM host

Cancel Active Tasks

- A complaint by *many* customers for a *long* time was that an entire DFSMSHsm address space had to be restarted whenever a single task was hung – generally due to tape problems
- **OA09344** - Enables cancellation of active DFSMSHsm *data movement* tasks
 - August 2005 ; All releases
- **QUERY ACTIVE (TCBADDRESS)**
 - Requests only messages directly related to specific data movement activities be listed along with extra information allowing cancellation of the active task
- **CANCEL TCBADDRESS()** or **SASINDEX()**
 - **TCBADDRESS** - Address of TCB returned from Q ACTIVE
 - **SASINDEX** - Index to be used when canceling an ABARS task
 - Can be done instead of canceling the ABARS started task

SMF Consolidation Processing

- Specify **DDCONS(NO)** on SMF parameters to avoid DD name consolidation during shutdown
 - DDCONS is specified in SMFPRMnn parmlib member
 - See *MVS Initialization and Tuning Reference* for more information
- DFSMShsm is a started task with thousands of DD name allocations
- DFSMShsm shutdown may be delayed up to 45 minutes if consolidation is performed
- SMF Type 30 records are a bit longer

Avoid LOG Overhead

- Use **HOLD LOG** to avoid DFSMSHsm logging overhead
 - Command can be added to PARMLIB
- Turns off writing to the LOGX/LOGY data sets
- Information available elsewhere, such as FSR records in SMF, Activity Logs, PDA trace data
- *Some ISV products require the LOGX/LOGY data sets as input*
- Reduces DFSMSHsm overhead activity (approximately 7% reduction)

DFSMSHsm PARMLIB

- PARMLIB **ONLYIF HSMHOST(hostid)** statement
 - Conditional processing based on Host identifier
 - Specify only the 1-character hostid, not the Y or N primary-host identifier
 - Single command following ONLYIF executed based on match on **HOST=x** parameter in startup procedure
 - Useful for sharing ARCCMDnn across systems
- Example
 - ONLYIF HSMHOST(1)
 - SETSYS PRIMARYSPMGMTSTART(400 730)
 - ONLYIF HSMHOST(2)
 - SETSYS PRIMARYSPMGMTSTART(900 1100)

DFSMSHsm PARMLIB *(cont)*

- SPECIFY EVERYTHING!!!
- Defaults *may* change with maintenance and new releases
 - This is highly avoided
- Improved change control if all commands appear in ARCCMDnn
- One-stop setup verification

DFSMSHsm PROCLIB

- Put **PDA=YES** in DFSMSHsm PROCLIB member
 - This allows tracing to start earlier during DFSMSHsm startup

Just For Fun

- Penalize a user who continuously Recalls hundreds/thousands of data sets on a frequent basis by periodically moving all their requests to the bottom of the queues:

ALTERPRI USERID(*anyuser*) LOW

Summary

- Improve performance
- Work smarter
- Exploit new functions
- Exploit technology

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